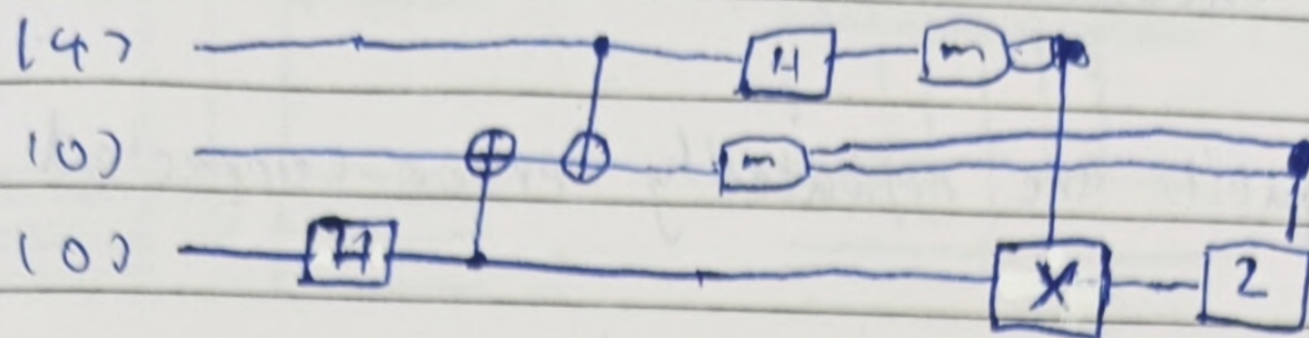


* Fault Tolerance

Consider a circuit,



Problems that occur → State Initializations

→ Unitary Gates

→ Measurements

→ Qubit Storage

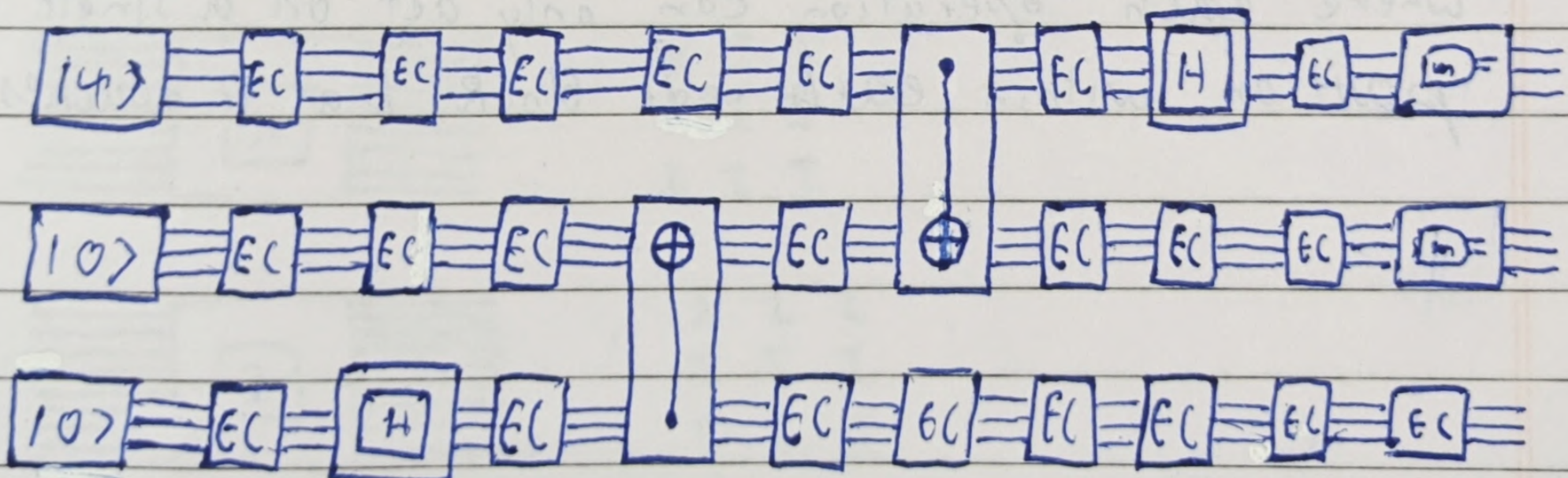
↳ Fault Tolerant

- To analyze fault-tolerant implementations of quantum circuits, we require a precise mathematical model

↓
A noise model

- As an example, one of the noise model says, different errors occur with different probabilities, independently, with no correlation → Independent Stochastic Noise Model

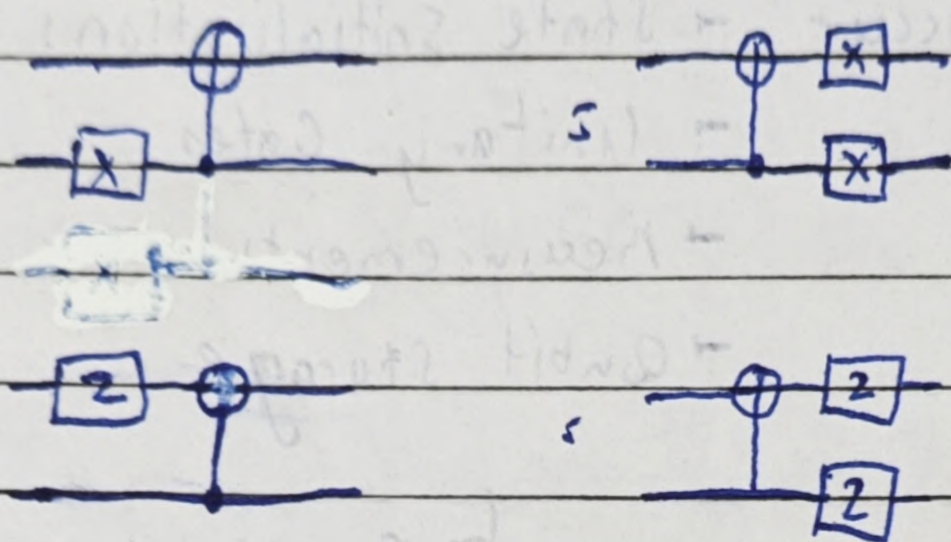
→ Fault Tolerance Implementation



EC = Error Correction

- State preparations, unitary gates, and measurements are performed by gadgets
- Qubits are encoded using QEC
- Encoded qubits are repeatedly error corrected

* Error Propagation



→ Even when CNOT is perfect, it can propagate errors

↳ Error Propagation

- This can cause correlated errors on two qubits

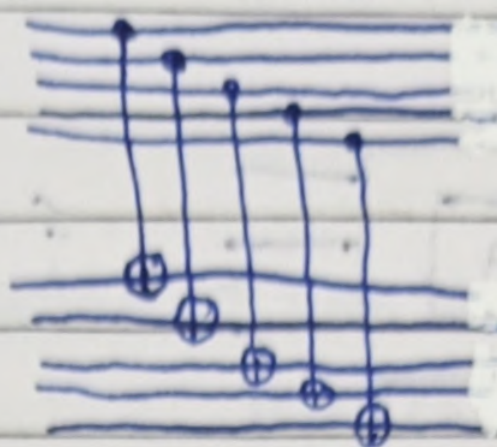
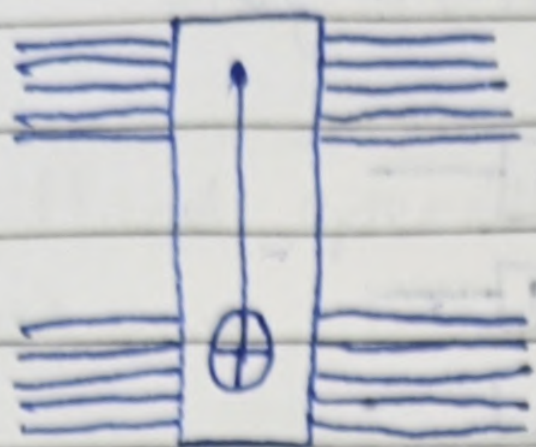
↓

Two-qubit gates can also be faulty causing correlated errors

* Transversal Gate Implementations

- It mitigates error propagation
- The gadgets must be a tensor product of operations, where each operation can only act on a single qubit position within each code block that it touches

- Examples of Transversal Gate Implementations
- CNOT gates for CSS code



→ Can't propagate error within the same code block

- Pauli gates for stabilizer codes

Any Pauli gate can be implemented

- 9-qubit Shor code

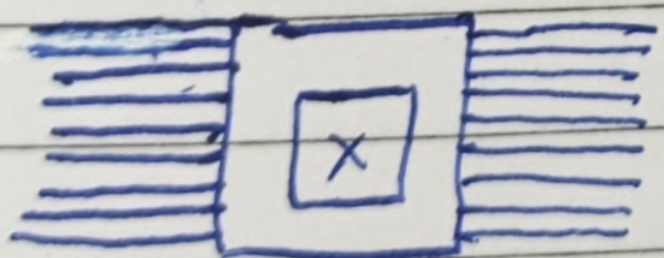
$$|0\rangle \mapsto \frac{1}{\sqrt{2}} (|1000\rangle + |1111\rangle) \otimes (|1000\rangle + |1111\rangle) \otimes (|1000\rangle + |1111\rangle)$$

$$|1\rangle \mapsto \frac{1}{\sqrt{2}} (|1000\rangle - |1111\rangle) \otimes (|1000\rangle - |1111\rangle) \otimes (|1000\rangle - |1111\rangle)$$

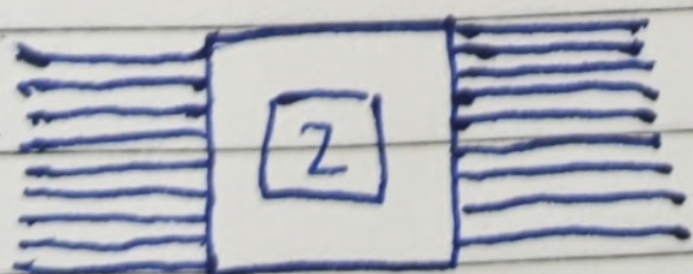
$$X \text{ gate} = Z \otimes I \otimes I \otimes Z \otimes I \otimes I \otimes Z \otimes I \otimes I$$

$$Z \text{ gate} = X \otimes X \otimes X \otimes I \otimes I \otimes I \otimes I \otimes I \otimes I$$

- 3x3 Surface code



$$\begin{matrix} X & I & I \\ X & I & I \\ X & I & I \end{matrix}$$



$$\begin{matrix} Z & Z & Z \\ I & I & I \\ I & I & I \end{matrix}$$



↓
we intentionally propagate errors
to classically detect them

→ Only works for CSS codes

- Knill error correction uses teleportation, it can correct arbitrary stabilizer codes

* Threshold Theorem

→ A quantum circuit having size N can be implemented with high accuracy by a noisy quantum circuit, provided that probability of error at each location in the noisy circuit is below a fixed, non-zero threshold value $p_{th} > 0$.

→ Size of noisy circuit scales as $O(N \log^c(N))$ for a positive constant c

→ p_{th} is Independent of N

* Fault Tolerant Error Correction

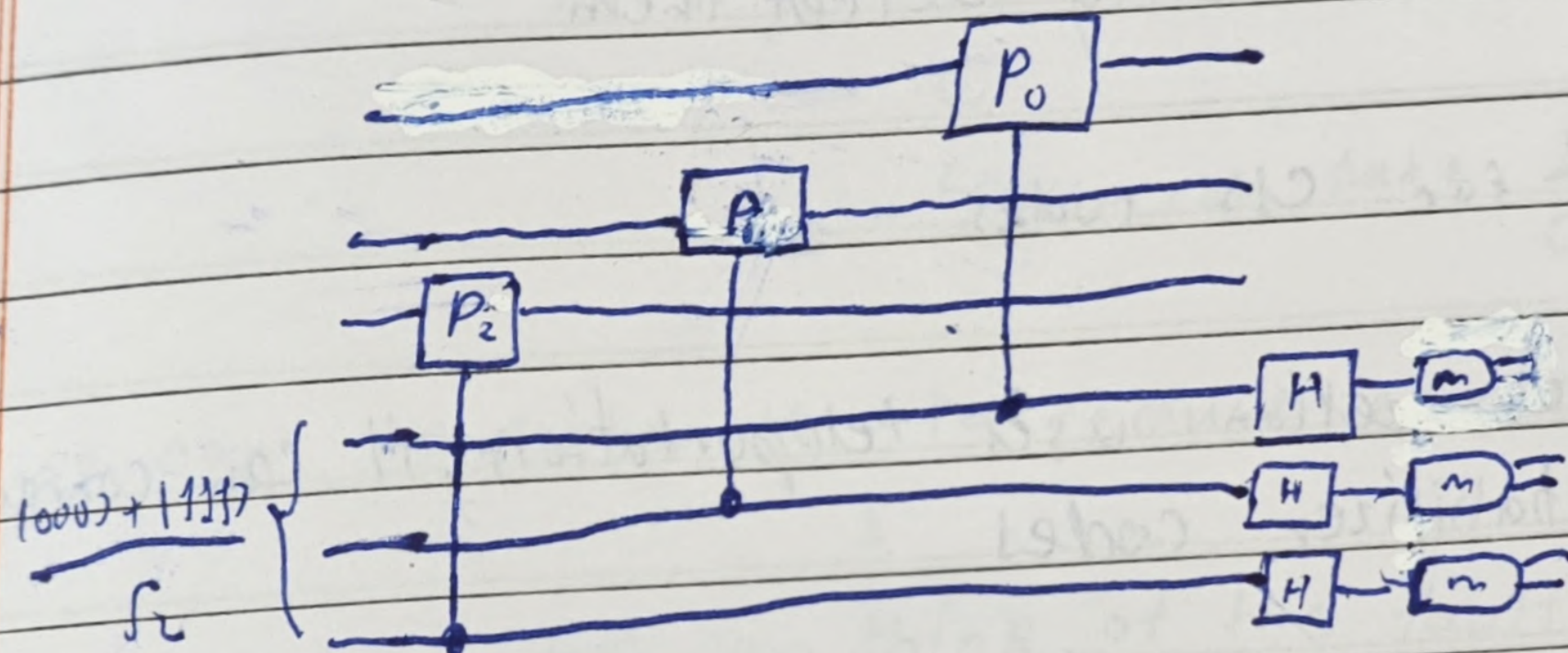
→ Errors could occur in QEC circuit

→ we need to prevent / correct those errors

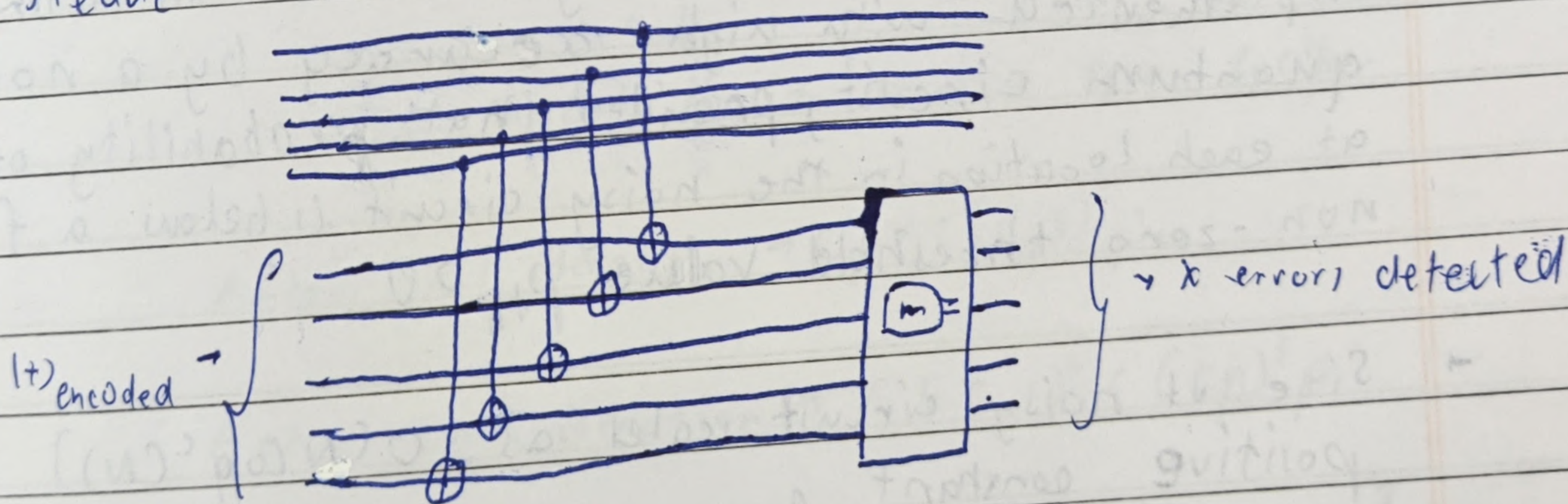
→ Shor Error Correction

we use cat state $\rightarrow \frac{1}{\sqrt{2}} |0^n\rangle + \frac{1}{\sqrt{2}} |1^n\rangle$

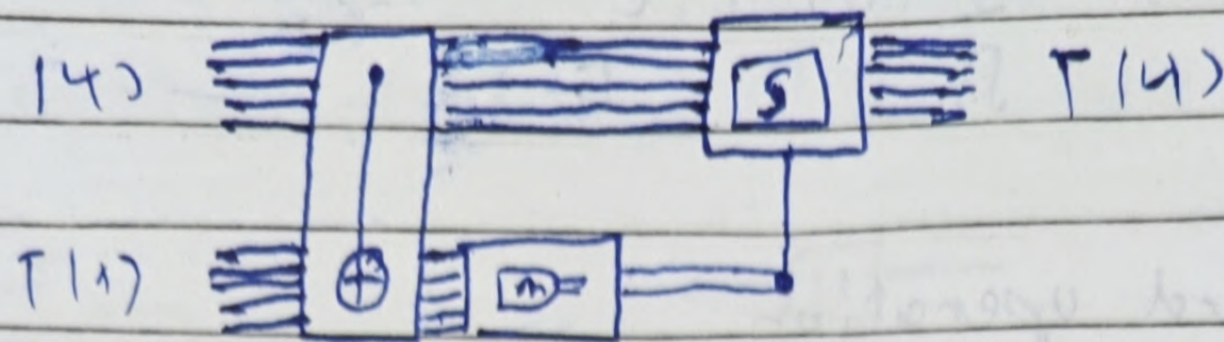
For $n=3$



→ Steane Error Correction

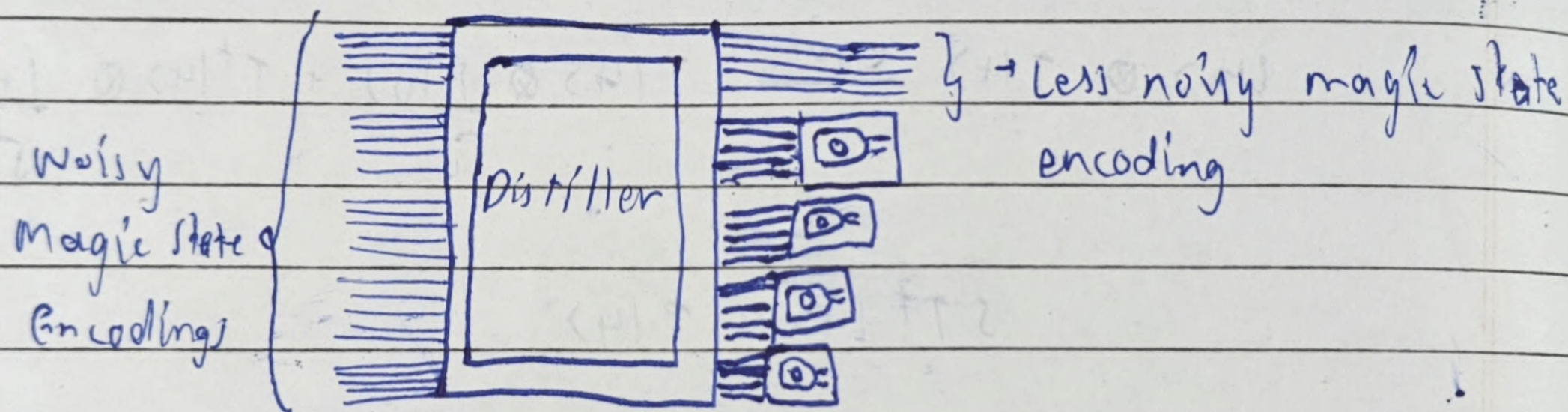


→ Fault Tolerant Gadgets from magic states



- we assume to have gadget of S gate
- To prepare $T|\psi\rangle$, we approximate it, as we don't have T gate
- we actually don't need $T|\psi\rangle$, we need its encoded state, hence we can approximate it too

→ Magic State Distillation



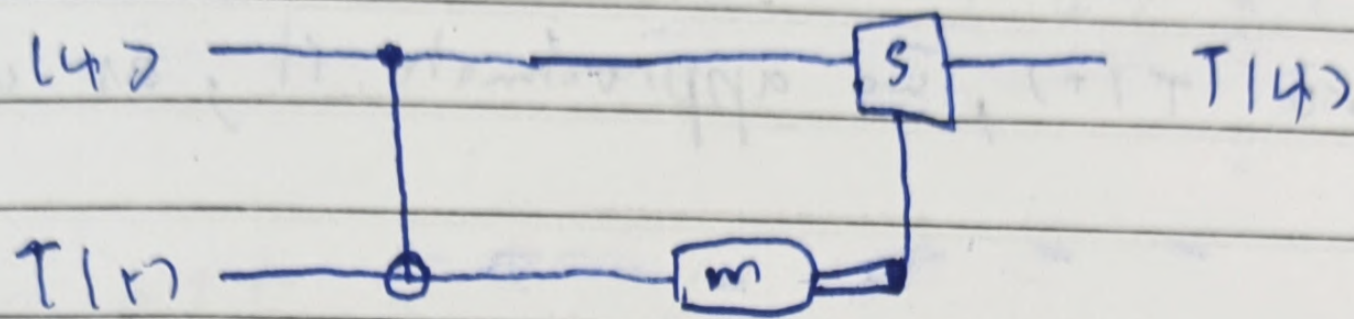
↳ If any of the outcomes is 1, the process has failed and we need to restart it

- we can run an encoding circuit for a stabilizer code in reverse to build the distiller

$$T|+\rangle = \frac{1}{\sqrt{2}}|0\rangle + \frac{e^{i\pi/4}}{\sqrt{2}}|1\rangle$$

→ S = Clifford operation

→ T = Not a Clifford operation + Can't be implemented using $H, S, CNOT$ only



→ Quantum Gate Teleportation

→ T gate is implemented on $|psi\rangle$

$T|+\rangle \rightarrow$ Magic State

$$|psi\rangle \otimes T|+\rangle \xrightarrow{CNOT} T|psi\rangle \otimes \frac{1}{\sqrt{2}}|0\rangle + T^\dagger|psi\rangle \otimes \frac{1+i}{\sqrt{2}}|1\rangle$$

$$S T^\dagger |psi\rangle = T|psi\rangle$$

→ Measurement outcome = 0

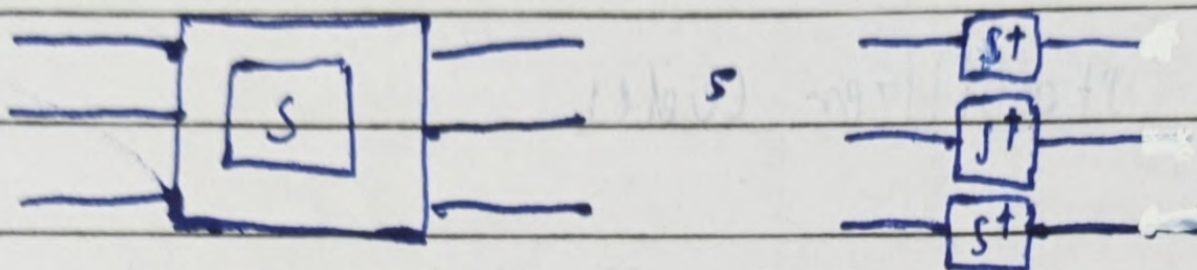
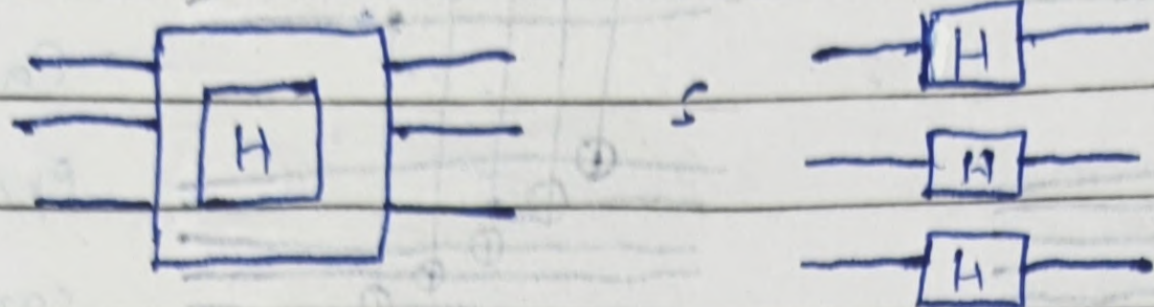
Posterior state = $T|psi\rangle$

→ Measurement outcome = 1

Posterior state = $T^\dagger|psi\rangle \xrightarrow{S} T|psi\rangle$

→ Gates for 7-qubit Steane code

- 7-qubit Steane code, every color code allows implementation of every clifford gate



→ Non-Universality of Transversal Gates

→ Eastin-Knill Theorem

- For any QEC code with distance ≥ 2 , the set of logical gates that can be implemented transversally generates a set of operations that is discrete, and therefore not universal

It is never possible, for any non-trivial QEC code, to implement a universal set of gates transversally

* Magic States

→ Implementing Gates from Magic states

$$S = \begin{bmatrix} 1 & 0 \\ 0 & i \end{bmatrix}, \quad \begin{bmatrix} 1 & 0 \\ 0 & e^{i\pi/2} \end{bmatrix}$$

$$T = \begin{bmatrix} 1 & 0 \\ 0 & \frac{1+i}{\sqrt{2}} \end{bmatrix}, \quad \begin{bmatrix} 1 & 0 \\ 0 & e^{i\pi/4} \end{bmatrix}$$

→ Say, we ~~use~~ ^{use} 7-qubit Steane code

→ 2 errors need to occur in the same code block to pass detection

$$P_{\text{error}} \rightarrow p$$

$$P_{\text{error using 7-qubit Steane code}} \rightarrow Cp^2$$

C → Fixed positive real number, depends on code and gadgets we use (not N)

$$\text{If } p < \frac{1}{C} \rightarrow \text{Reduction in error}$$

$$\hookrightarrow \text{we can take } \frac{1}{C} \text{ as } p_{th} \rightarrow \frac{1}{C} p_{th}$$

→ we can concatenate to reduce error

↓

we can think of the fault-tolerant implementation of the original circuit as if it were any other circuit

↓

we repeat this to decrease error further

$$\text{After 1st iteration} \rightarrow Cp^2 = (Cp) p$$

$$\text{After 2nd iteration} \rightarrow C(Cp^2)^2 = (Cp)^3 p$$

$$\text{After 3rd iteration} \rightarrow C((Cp)^3 p)^2 = (Cp)^7 p$$

↓

$$\text{After } k\text{th iteration} \rightarrow (Cp)^{2^k - 1} p \rightarrow \text{Increases doubly exponentially}$$

→ size increases only singly exponentially